

MONOCOTS

Defined by their unique internal anatomy, monocots include grasses and palms, plus lilies, orchids, and many other ornamental plants.

Early in the evolution of flowering plants, their family tree developed two major branches. A smaller but still substantial branch became the monocots, while the larger branch became the eudicots. Monocots get their name from their seeds, which have a single pre-packed seed-leaf, or cotyledon. Apart from this, there is no foolproof way of recognizing a monocot, although there are strong clues. Most have long, narrow leaves with parallel veins, and the flower parts (petals, stamens, and so on) are arranged in threes or multiples of three. The pollen grains have a single opening, unlike those of eudicots which have three. In many monocot flowers, such as tulips, the sepals and petals are almost identical and are known as tepals, but many magnoliids share this feature.

Below ground, monocots usually have a cluster of fibrous roots, instead of a main taproot with smaller roots leading off it. Another key

characteristic – revealed under a microscope – involves the structure of their stems. Monocots have scattered bundles of vascular tissue – the specialized cells that carry water and sap – unlike eudicots, which have theirs in concentric rings. This makes monocot stems more flexible than those of a typical eudicot, but it also makes it harder for them to evolve into trees. Tree-like monocots, which principally include palms, have a very different way of growing to typical broadleaved trees and conifers. Monocot trunks grow taller, but not thicker, and are usually topped by a single rosette of leaves.

SURVIVAL STRATEGIES

Smaller monocots vary greatly, from climbers to aquatic plants that survive harsh periods as underground storage organs, such as bulbs or tubers. Grasses thrive on the grazing effects of animals, and are the only plant family that forms an entire habitat: grassland. In the tropics, many monocots are epiphytic, growing high up in trees. This group includes bromeliads, and many orchids – by far the largest monocot family, with over 25,000 species worldwide.

DIVISION	ANGIOSPERMAE
CLADE	MONOCOTS
ORDERS	11
FAMILIES	77
SPECIES	60,000

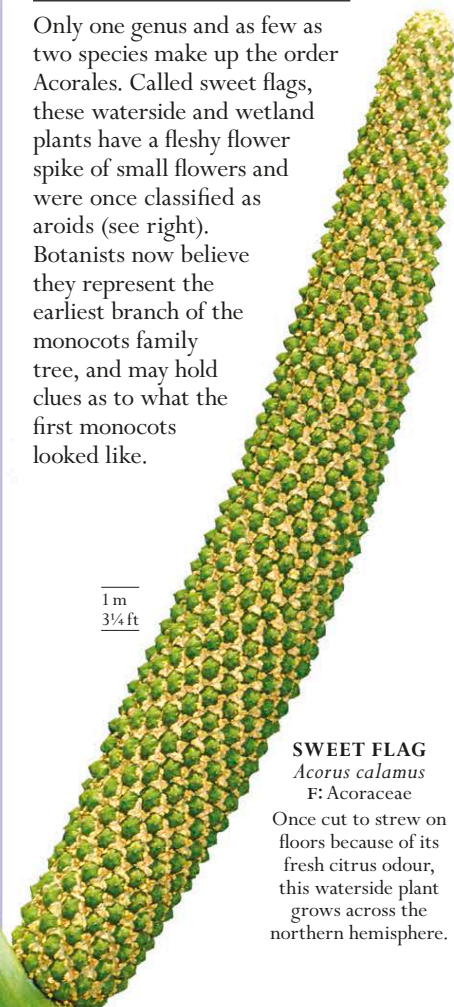
DEBATE

DID MONOCOTS HAVE AQUATIC ORIGINS?

Living monocots include many freshwater plants, and also some of the few flowering plants that live in the sea. According to a longstanding theory, monocots may have originally evolved in fresh water, before diversifying and taking up life on land. This would account for the long, thin leaves found in many species, and also the internal structure of their stems (see left). Having spread to land, some terrestrial monocots then came full circle and evolved into aquatic forms – duckweeds, for example, originated in this way.

ACORALES

Only one genus and as few as two species make up the order Acorales. Called sweet flags, these waterside and wetland plants have a fleshy flower spike of small flowers and were once classified as aroids (see right). Botanists now believe they represent the earliest branch of the monocots family tree, and may hold clues as to what the first monocots looked like.



SWEET FLAG
Acorus calamus
F: Acoraceae

Once cut to strew on floors because of its fresh citrus odour, this waterside plant grows across the northern hemisphere.

ALISMATALES

This order includes many common aquatic plants as well as the mainly land-based aroid family (Araceae). Aroids, which can be dramatic in appearance, have a distinctive reproductive anatomy consisting of a fleshy spike of tiny flowers called a spadix and a leaf-like surround called a spathe. The other families in the order include many freshwater species, as well as several families of seagrasses.

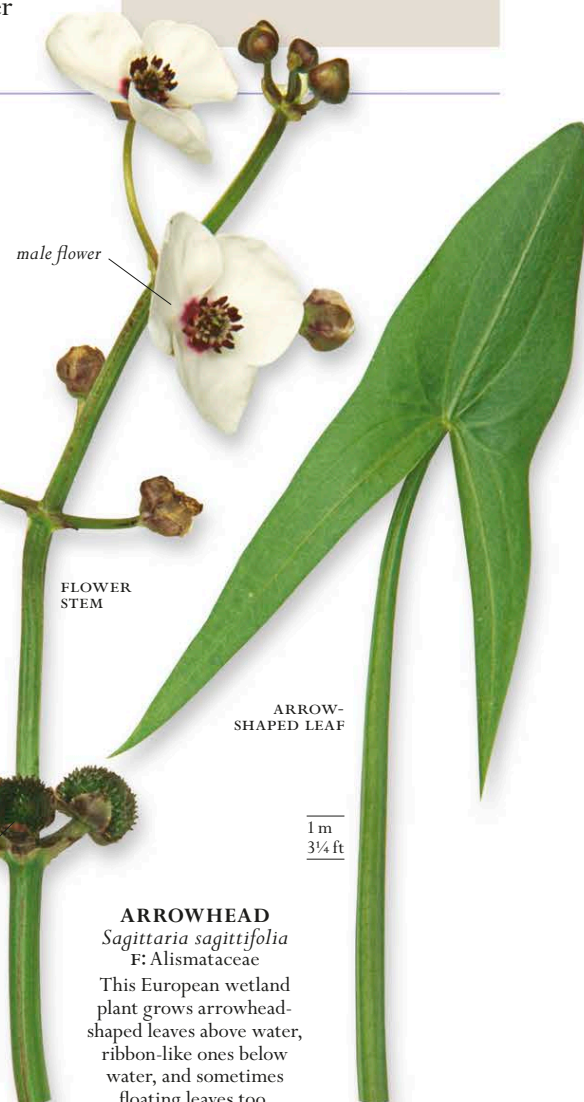
WATER-PLANTAIN
Alisma plantago-aquatica
F: Alismataceae

Found in waterside habitats, this plant is common across Europe, Asia, and N. Africa. It has white, pinkish, or pale purple flowers that last only a day.



WATER HAWTHORN
Aponogeton distachyos
F: Aponogetonaceae

Native to South Africa, this plant has become widely naturalized elsewhere. Its vanilla-scented flowers open just above the water surface.



ARROWHEAD
Sagittaria sagittifolia
F: Alismataceae

This European wetland plant grows arrowhead-shaped leaves above water, ribbon-like ones below water, and sometimes floating leaves too.